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Education

- Ph.D. in Biomedical Science and Engineering, Konkuk University, 2021 – Present (expected Feb. 2027) — Advisor: Prof. Dae-Jin Yun; GPA 4.31/4.5
- M.S. in Biomedical Science and Engineering, Konkuk University, 2019 – 2021 — Thesis: Structural and Functional Analysis of HOS15, a Substrate Receptor of CRL4 in Cold Stress Signaling; GPA 4.45/4.5
- B.S. in Biomedical Science and Engineering, Konkuk University, 2013 – 2019 — Thesis: Functional Studies of SUMOylation of Histone Deacetylase in Plants; GPA 3.07/4.5

Research Focus

My research investigates the epigenetic regulation of plant stress adaptation, focusing on the linker histone H1 and chromatin remodeling in *Arabidopsis thaliana*. Building on earlier work on HOS15–PWR–HDAC complexes in cold stress signaling, I study how histone H1 controls chromatin accessibility and transcriptional reprogramming under abiotic stress, using CRISPR-based genetics, ChIP-seq, RNA-seq, and confocal imaging.

Research Projects

- Global Plant Stress Research Center (NRF), 2024.08–2030.04 — Participating Researcher
- Genome-wide chromatin reprogramming and transcriptome regulation in plant environmental stress adaptation (NRF), 2026.03–2031.02 — Participating Researcher
- Chromatin reconfiguration and transcriptional control for plant stress adaptation (NRF), 2022.03–2027.02 — Co-participating Researcher

Publications (SCIE/ESCI)

- 1 Ali A, Zareen S, Bader ZE, Park J, Khan IU, Park K, Albakri N, **Bae MJ**, Bressan RA, Pardo JM, Yun DJ, Xu ZY. "A GPS2-like protein interacts with HOS15 and HDA6 to form a repressor complex that regulates ABA signaling and drought adaptation in Arabidopsis." *Plant Communications*, 2026.
- 2 **Bae MJ**, Zareen S, Yun H, Ali A, Bader ZE, Park K, Albakri N, Jung YJ, Yun DJ, Park J. "Unveiling the epigenetic and stress-responsive function of histone H1 in plants." *Plant Signaling & Behavior*, 2026.
- 3 Albakri N, Bader ZE, Park KS, **Bae MJ**, Zareen S, Yun H, Pradhan S, Yun DJ, Park J. "HOS15 Modulates Alternative Splicing Homeostasis Under Cold Stress in Arabidopsis Thaliana." *Journal of Plant Biology*, 2025.
- 4 Park K, Lim CJ, Albakri N, **Bae MJ**, Park J. "HOS15 Contributes to Thermotolerance Through Destabilization of HD2C in Arabidopsis." *Journal of Plant Biology*, 2025.
- 5 Zareen S, Ali A, **Bae MJ**, Albakri N, Park K, Yun H, Yun DJ, Park J. "HOS15 together with PWR–HDA9 positively regulates dark-induced senescence in Arabidopsis." *Plant Signaling & Behavior*, 2025.
- 6 Bader ZE, **Bae MJ**, Ali A, Park J, Baek D, Yun DJ. "GIGANTEA-ENHANCED EM LEVEL complex initiates drought escape response via dual function of ABA synthesis and flowering promotion." *Plant Signaling & Behavior*, 2023.
- 7 Lim CJ, Park KS, Ali A, Park J, Ryou SM, Shen M, Khan HA, Bader ZE, Zareen S, **Bae MJ**, Choi JH, Xu ZY, Pardo JM, Yun DJ. "Negative regulation of floral transition in Arabidopsis by HOS15-PWR-HDA9 complex." *Frontiers in Plant Science*, 2023.
- 8 Lim CJ, Park J, Shen M, Park HJ, Cheong MS, Park K, Baek D, **Bae MJ**, Ali A, Jan M, Lee SY, Lee BH, Kim WY, Pardo JM, Yun DJ. "The Histone-Modifying Complex PWR/HOS15/HD2C Epigenetically Regulates Cold Tolerance." *Plant Physiology*, 2020.

Manuscripts in preparation (targeting PNAS): Mechanisms of Histone H1 degradation under cold stress; HOS15-dimerization promotes HD2C degradation during Cold Stress.

Patents

- Vector for improving productivity of cellulose and transformed mouseear cress by the same — KR 10-2023-0113597
- Vector for improving productivity and storage capacity of vegetable oil and transformed plant by the same — KR 10-2024-0168464
- Vector for improving productivity of cellulose and transformed mouseear cress by the same — KR 10-2024-0168432
- Sucrose-accumulating plant and method for preparing the same — KR 10-2025-0104961
- Ethanol production and tolerance enhanced yeast strain and method for producing ethanol using the same — KR 10-2025-0104960
- Recombinant Kluyveromyces marxianus strain with engineered xylose metabolic pathway and method for producing ethanol using the same — KR 10-2025-0104960

Awards and Honors

- Excellent Poster Award, Korean Society of Plant Biologists (KSPB), Korea, Nov. 2021
- HOPE Fellow, HOPE Meetings with Nobel Laureates, Japan Society for the Promotion of Science (JSPS), Japan, Mar. 2024

Conference Presentations

- 2026 GPSRC International Conference, Konkuk University, Korea, May 2026
- 2026 Winter Conference of the Korean Society of Plant Biologists, Korea University, Korea, Feb. 2026 — Poster
- 2nd Global Plant Stress Research Center (GPSRC) & 2025 KSMCB Plant Molecular Biology International Symposium, Seoul, Korea, Aug. 2025 — Organizer
- 15th HOPE Meeting, Kyoto, Japan, Feb. 2024 — Poster
- 2024 International Conference of the Korean Society of Plant Biologists, Seogwipo-si, Korea, Nov. 2024 — Oral Presentation
- 2022 International Conference of the Korean Society of Plant Biologists, Busan, Korea, Nov. 2022 — Poster
- 2021 International Conference of the Korean Society of Plant Biologists, Online, Nov. 2021 — Poster
- 2020 International Conference of the Korean Society of Plant Biologists, Online, Nov. 2020 — Poster
- 2019 Annual Conference of the Korean Society of Plant Biologists, Buyeo, Korea, Nov. 2019 — Poster
- 2018 International Conference of the Korean Society of Plant Biologists, Busan, Korea, Nov. 2018 — Poster

Skills

- **Plant molecular biology:** Gateway cloning, Gibson assembly, vector construction, transgenic line generation, Arabidopsis genetics and phenotypic analysis
- **Epigenetics & chromatin biology:** histone biology, chromatin remodeling, nucleosome organization, ChIP-seq, open-chromatin/accessibility analysis
- **Omics & data analysis:** RNA-seq, LC-MS proteomics, multi-omics factor analysis (MOFA/MOFA2), alternative splicing analysis
- **Protein & stress biology:** ubiquitin–proteasome pathway, E3 ligase complexes, protein stability, cold/drought/thermotolerance and senescence signaling
- **Software/tools:** R (MOFA/MOFA2), HOMER, GraphPad Prism, confocal/immunofluorescence imaging analysis